



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
15-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]

PAPER-2

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

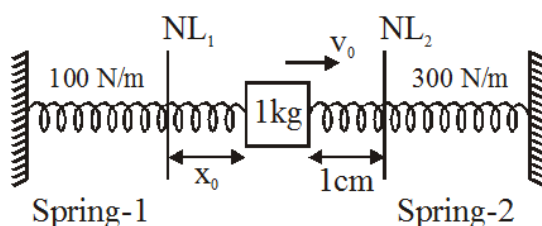
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

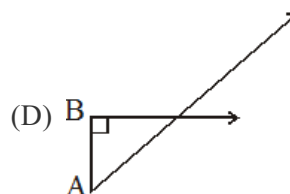
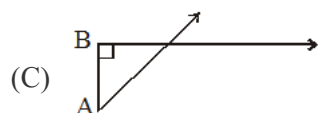
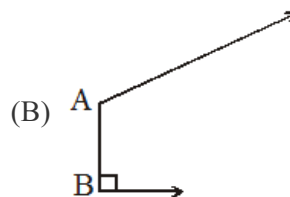
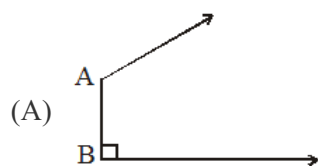
1. Block is in equilibrium in the situation shown then (Here NL_1 and NL_2 are the natural length of springs respectively)



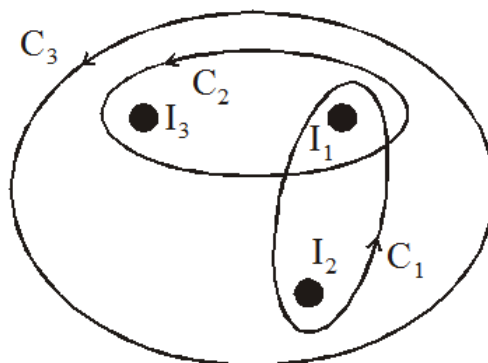
- (A) Value of x_0 is 3cm.
- (B) Time period of SHM is 0.314 sec.
- (C) If kinetic energy of block is minimum when spring-1 is in its natural length then $v_0 = 0.6$ m/s.
- (D) If kinetic energy of block is minimum when spring-2 is in its natural length then $v_0 = 0.2$ m/s.

Space for Rough Work

2. Each of the following figures shows electric field vectors at two points A and B in an electric field. In which figure of figures can the illustrated field be created by a single point charge?



3. Three infinitely-long conductors carrying currents I_1 , I_2 and I_3 lie perpendicular to the plane of the paper as shown below. If the value of integral $\oint_C \vec{B} \cdot d\vec{\ell}$ for the loops C_1 , C_2 and C_3 are $2\mu_0$, $4\mu_0$ and μ_0 in SI units respectively, then



(A) $I_1 = 5\text{A}$ out of the paper

(B) $I_2 = 3\text{A}$ into the paper

(C) $I_3 = 0$

(D) $I_3 = 1\text{A}$ into the paper

Space for Rough Work

4. In the circuit shown, two parallel plate capacitors of capacitance C and $2C$ consisting of thin plates of identical dimension. They are connected with an ideal battery of terminal voltage V as shown in figure I. Now the capacitor smaller in volume is completely inserted into the other capacitor larger in volume and kept in symmetric position and then connection polarities of the plates are reversed as shown in figure-II. Identity correct statements.

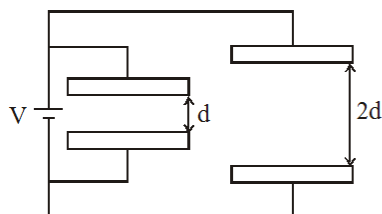


Figure-I

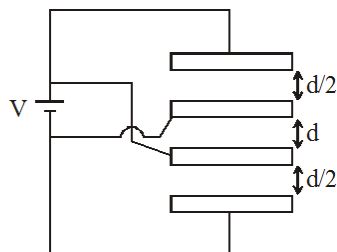
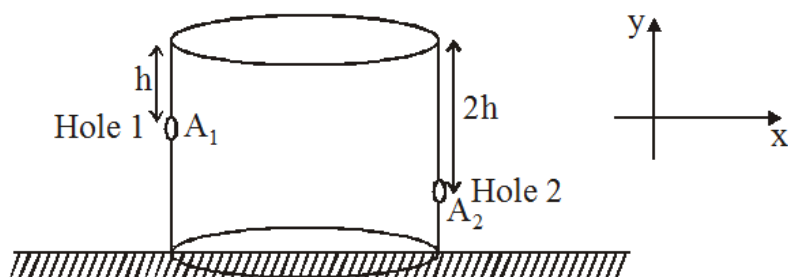


Figure-II

- (A) Capacitance of the new system is $10C$. (B) Capacitance of the new system is $7C$.
 (C) Charge flown through the battery is $10CV$. (D) Charge flown through the battery is $7CV$.
5. Choose the correct statements (s) :
- (A) A neutrino is emitted in positron decay and an antineutrino is emitted in electron decay.
 (B) Electron capture, occurs when a parent nucleus capture one of its own orbital electrons and emits a neutrino.
 (C) In carbon dating we use ^{14}C instead of ^{11}C because ^{14}C half-life is smaller then ^{11}C .
 (D) Nucleus having higher binding energy must be more stable irrespective of mass number of nucleus.

Space for Rough Work

6. A cylinder which was initially completely filled with liquid having two holes as shown in figure is on rough horizontal surface then choose correct statement(s) (Until level of liquid fall upto 1st hole)



- (A) If $A_1 = 2A_2$ then initially friction does not act on cylinder.
 (B) If $A_1 = A_2$ friction on cylinder remain unchanged all the time.
 (C) If $A_1 = 2A_2$ then after some time friction on cylinder acts towards negative x -direction.
 (D) If $A_1 = A_2$, magnitude of friction first decrease then increases.
7. If the tension in a stretched string fixed at both ends is changed by 21%, the fundamental frequency is found to increase by 15 Hz then the
- (A) Original fundamental frequency is 150 Hz.
 (B) Velocity of propagation of the transverse wave along the string changes by 5%.
 (C) Velocity of propagation of the transverse wave along the string changes by 10%.
 (D) Fundamental wavelength on the string does not change.
8. What can be the possible values of angle of prism (A) so that whatever be the angle of incidence, all the rays will emerge from other side i.e. there is no total internal refraction on second surface ? Refractive index of prism is $\mu = \frac{5}{4}$. (Consider cases in which formula of prism are applicable)
- (A) 15° (B) 45° (C) 30° (D) 60°

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

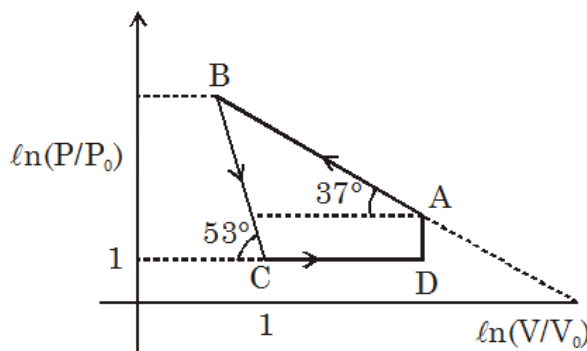
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.

Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)

Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

For a thermodynamic cycle ABCD on a non-linear polyatomic gas, we draw $\ln\left(\frac{P}{P_0}\right)$ versus $\ln\left(\frac{V}{V_0}\right)$ diagram.



Match the process.

List-I		List-II	
(I)	AB	(P)	Polytropic $PV^m = \text{constant}$. ($m \neq \gamma$ or 1 or 0)
(II)	BC	(Q)	Adiabatic
(III)	CD	(R)	Isochoric
(IV)	DA	(S)	Isobaric
		(T)	Isothermal
		(U)	$P = a + bV$

(A) I $\rightarrow$ P; II $\rightarrow$ U; III $\rightarrow$ R; IV $\rightarrow$ S

(B) I $\rightarrow$ P; II $\rightarrow$ P; III $\rightarrow$ T; IV $\rightarrow$ R

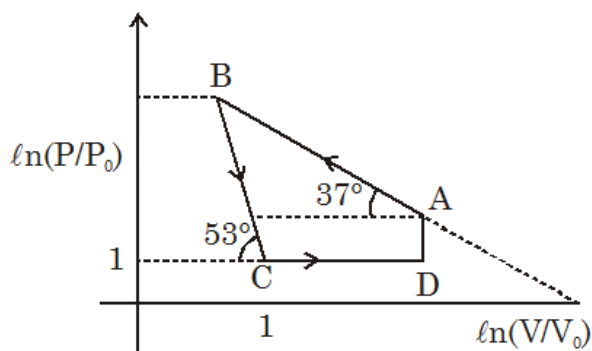
(C) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ R

(D) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ R

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

For a thermodynamic cycle ABCD on a non-linear polyatomic gas, we draw $\ln \left(\frac{P}{P_0} \right)$ versus $\ln \left(\frac{V}{V_0} \right)$ diagram.



Match the process.

List-I		List-II	
(I)	W_{A-B}	(P)	Positive
(II)	Q_{B-C}	(Q)	Negative
(III)	ΔU_{CD}	(R)	Zero
(IV)	Q_{D-A}	(S)	Can't be said
		(T)	Initially positive and negative after that
		(U)	Initially negative and positive after that

(A) I $\rightarrow$ T; II $\rightarrow$ P; III $\rightarrow$ P; IV $\rightarrow$ P

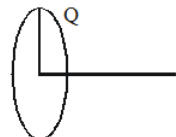
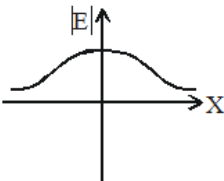
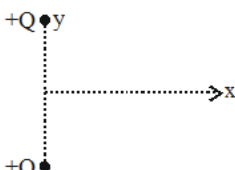
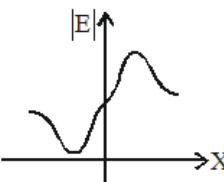
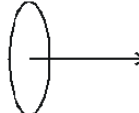
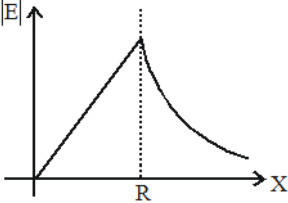
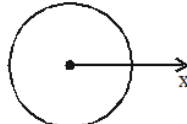
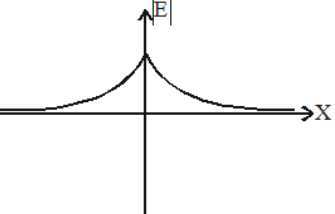
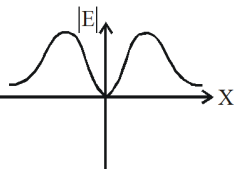
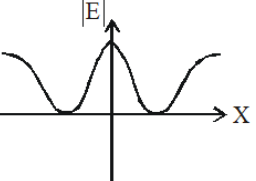
(B) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ P; IV $\rightarrow$ P

(C) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ R

(D) I $\rightarrow$ S; II $\rightarrow$ Q; III $\rightarrow$ Q; IV $\rightarrow$ P

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.
For the system of charges and electric field at the given axis.

List-I		List-II	
(I)	 Uniformly positively charged ring in yz plane with axis along x axis	(P)	
(II)	 Two identical positively charged article placed on y axis symmetrical to x axis	(Q)	
(III)	 Uniformly positively charged on conducting disc in yz plane with axis along x axis.	(R)	
(IV)	 Uniformly positively charged (charge distribute throughout its volume) non conducting sphere with diameter axis along x-axis.	(S)	
		(T)	
		(U)	

(A) I $\rightarrow$ U; II $\rightarrow$ U; III $\rightarrow$ P; IV $\rightarrow$ R

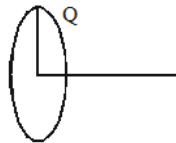
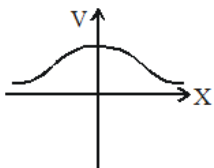
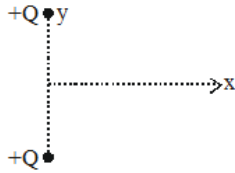
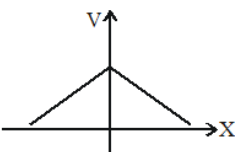
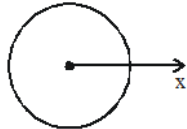
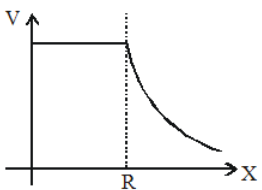
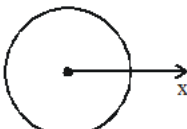
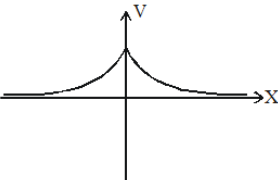
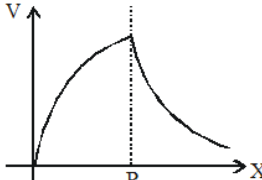
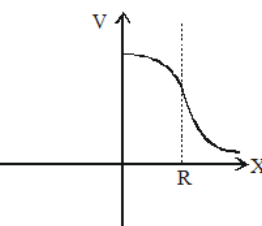
(B) I $\rightarrow$ T; II $\rightarrow$ T; III $\rightarrow$ P; IV $\rightarrow$ R

(C) I $\rightarrow$ T; II $\rightarrow$ T; III $\rightarrow$ S; IV $\rightarrow$ R

(D) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ T

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

For system of charges and electric potential at given axis (Take : potential is zero at infinite)

List-I		List-II	
(I)	 Uniformly positively charged ring in yz plane with axis along x axis	(P)	
(II)	 Two identical positively charged article placed on y axis symmetrical to x axis	(Q)	
(III)	 Uniformly positively charged conducting sphere with diameter axis along x-axis.	(R)	
(IV)	 Uniformly positively charged (charge distribute throughout its volume) non conducting sphere with diameter axis along x-axis.	(S)	
		(T)	
		(U)	

(A) I $\rightarrow$ S; II $\rightarrow$ S; III $\rightarrow$ R; IV $\rightarrow$ T

(B) I $\rightarrow$ Q; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ T

(C) I $\rightarrow$ S; II $\rightarrow$ R; III $\rightarrow$ R; IV $\rightarrow$ U

(D) I $\rightarrow$ P; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ U

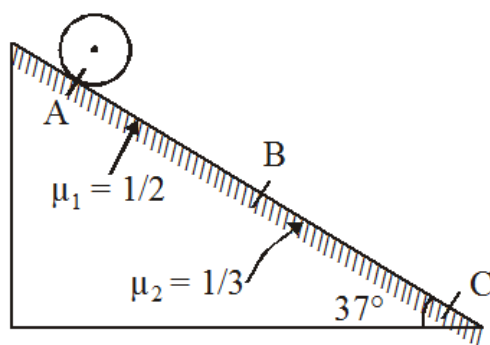
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

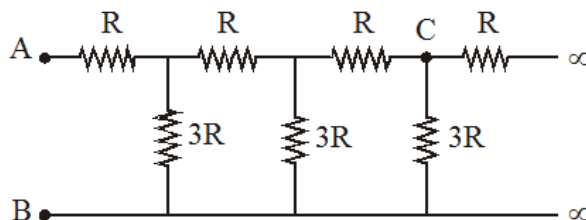
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. As shown in figure a disc is released from rest where coefficient of friction between disc & inclined plane for path AB is $\frac{1}{2}$ & for BC is $\frac{1}{3}$. If $AB = BC$ then find the ratio of time interval $\frac{\Delta t_{BC}}{\Delta t_{AB}}$.



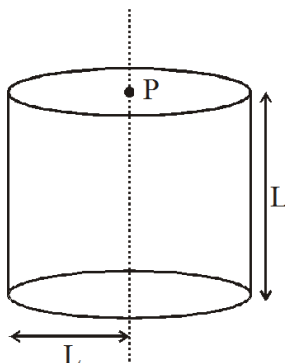
2. If potential difference between C & B is 20V then find potential difference between A & B (in volt).



Space for Rough Work

3. A thin hollow cylinder (opened at both ends) of radius and length both equal to L . This object carries a uniform surface-charge density σ . The electrostatic potential at the point P on the axis of the cylinder (as shown in the figure) comes out to be $V_P = \frac{\sigma L}{k\epsilon_0} \ln(1 + \sqrt{2})$. Find the value of k .

[You may use $\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{x^2 + a^2})$.]



4. In a Young's interference experiment, the two slits are separated by 0.150 mm and the incident light includes two wavelengths: $\lambda_1 = 540 \text{ nm}$ (green) and $\lambda_2 = 450 \text{ nm}$ (blue). The overlapping interference pattern are observed on a screen 1.40 m from the slits. Calculate the minimum distance (in cm) from the center of screen to a point where a bright fringe of the green light coincides with a bright fringe of the blue light.
5. Longest wavelength of lyman series for H atom is $\lambda_1 \text{ nm}$ & shortest wavelength of Balmer series is $\lambda_2 \text{ nm}$ then the value of $(\lambda_2 - \lambda_1)$ is : [Take $R = 1.1 \times 10^7 / \text{m}$.]
6. Calculate the wavelength (in $\text{\AA}$) of the x-ray emitting from a molybdenum target ($z = 42$), when an electron moves from L shell to K shell. (Take $R = 1.1 \times 10^7 / \text{m}$)

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

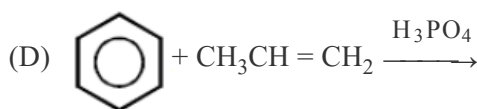
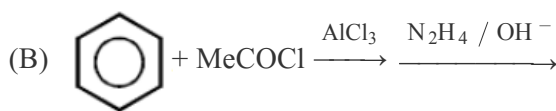
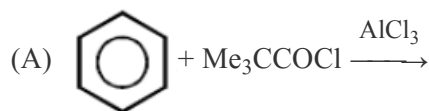
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

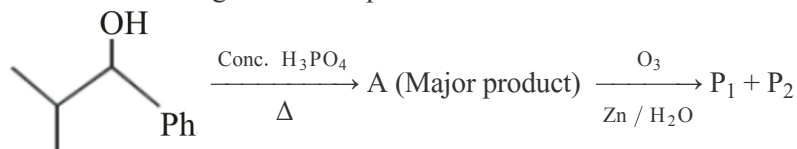
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Which of the following reaction(s) can make alkyl benzene ?



Space for Rough Work

2. Consider following reaction sequence :



P₁ undergoes aldol condensation while P₂ can not. Which of the following statement(s) is/are **INCORRECT** ?

- (A) NaHSO₃ can be used to distinguish P₁ and P₂.
- (B) P₂ reduces tollens reagent but can not reduce Fehling's solution.
- (C) P₁ and P₂ can undergo iodoform test.
- (D) Compound P₂ can give positive 2,4-DNP test but can not undergo Cannizzaro reaction.
3. Select incorrect statement :
- (A) Soaps are sodium or potassium salt of higher carboxylic acid like stearic acid
- (B) Penicilline is a sweetening agent
- (C) Monomer of Neoprene is isoprene
- (D) With one equivalent Na / liq. NH₃ propyne give propene and reaction is known as Birch reduction

Space for Rough Work

4. Incorrect statements is/are from the following -
- (A) Adsorption is exothermic in nature
 - (B) CaCl_2 causes coagulation of gold sol more easily than AlCl_3
 - (C) Surface tension of associated colloids increases with increase in concentration of their aqueous sols.
 - (D) With increase in concentration, conductance hydrosols of associated colloids increases more rapidly after CMC than before it.
5. For first order reaction with single reactant :
- (A) time taken for completion of 75% of reaction is twice that of half-life
 - (B) Rate constant of reaction increases linearly with temperature.
 - (C) The reaction may be either elementary or complex.
 - (D) Time taken for reaction to be 80% complete will not depend on initial concentration of reactant.
-

Space for Rough Work

6. Which of the following coloured spin paired complex(es) shows isomerism?
- (A) $[\text{Fe}(\text{en})_3] \text{Cl}_3$ (B) $[\text{Sc}(\text{H}_2\text{O})_6] \text{Cl}_3$
(C) $\text{Cu}_2[\text{Fe}(\text{CN})_6]$ (D) $\text{Na}_3[\text{Cr}(\text{ox})_3]$
7. In which of the following reactions, xenon fluorides do not give redox reaction?
- (A) $\text{XeF}_6 + \text{SiO}_2 \rightarrow$ (B) $\text{XeF}_4 + \text{H}_2\text{O} \rightarrow$
(C) $\text{XeF}_6 + \text{H}_2\text{O} \rightarrow$ (D) $\text{XeF}_2 + \text{H}_2\text{O} \rightarrow$
8. Calcination and roasting are carried out to :
- (A) convert ores into their respective metal oxide
(B) remove water of crystallisation
(C) melt the ore
(D) remove impurities into their respective volatile oxides
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.

Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)

Negative Marks : -1 In all other cases

9.

	List-I Conversion		List-II Reagents
(I)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2 \rightarrow \text{R}-\text{NH}_2$	(P)	$\text{NaNO}_2 / \text{HCl} / \Delta / \text{H}_3\text{O}^{\oplus}$
(II)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{R} \rightarrow \text{R}-\text{NH}_2$	(Q)	$\text{N}_3\text{H} / \text{H}_2\text{SO}_4 / \Delta$
(III)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH} \rightarrow \text{R}-\text{NH}_2$	(R)	$\text{KOH} / \text{Br}_2 / \Delta$
(IV)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}-\text{NH}_2 \rightarrow \text{R}-\text{NH}_2$	(S)	NaOH / Δ
		(T)	$\text{HgCl}_2 / \text{CS}_2 / \Delta$
		(U)	$\text{Hg}(\text{OAc})_2 / \text{NaBH}_4 / \text{H}_2\text{O}$

Which of the following having correct combination considering List-I and List-II

(A) II $\rightarrow$ T

(B) II $\rightarrow$ P

(C) III $\rightarrow$ Q

(D) IV $\rightarrow$ T

Space for Rough Work

10.

	List-I Conversion		List-II Reagents
(I)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2 \rightarrow \text{R}-\text{NH}_2$	(P)	$\text{NaNO}_2 / \text{HCl} / \Delta / \text{H}_3\text{O}^{\oplus}$
(II)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{R} \rightarrow \text{R}-\text{NH}_2$	(Q)	$\text{N}_3\text{H} / \text{H}_2\text{SO}_4 / \Delta$
(III)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH} \rightarrow \text{R}-\text{NH}_2$	(R)	$\text{KOH} / \text{Br}_2 / \Delta$
(IV)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}-\text{NH}_2 \rightarrow \text{R}-\text{NH}_2$	(S)	NaOH / Δ
		(T)	$\text{HgCl}_2 / \text{CS}_2 / \Delta$
		(U)	$\text{Hg}(\text{OAc})_2 / \text{NaBH}_4 / \text{H}_2\text{O}$

Which of the following having correct combination considering List-I and List-II

- (A) II $\rightarrow$ T (B) III $\rightarrow$ U (C) IV $\rightarrow$ S (D) I $\rightarrow$ R

11.

	List-I (Type of process)		List-II (Parameter change)
(I)	Isothermal free expansion of ideal gas	(P)	$\Delta S_{\text{system}} > 0$
(II)	$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g}) + \text{Heat}$	(Q)	$\Delta S_{\text{surr.}} = 0$
(III)	Condensation of water vapour at 1 atm	(R)	$\Delta S_{\text{system}} < 0$
(IV)	Adiabatic reversible compression of ideal gas	(S)	$\Delta S_{\text{surr.}} > 0$
		(T)	$W > 0$
		(U)	$\Delta H = 0$ or $\Delta H < 0$

Which of the following having correct combination considering List-I and List-II

- (A) I $\rightarrow$ P,Q,U (B) II $\rightarrow$ P,Q
(C) III $\rightarrow$ P,T (D) IV $\rightarrow$ Q,R

Space for Rough Work

12.

	List-I (Type of process)		List-II (Parameter change)
(I)	Isothermal free expansion of ideal gas	(P)	$\Delta S_{\text{system}} > 0$
(II)	$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g}) + \text{Heat}$	(Q)	$\Delta S_{\text{surr.}} = 0$
(III)	Condensation of water vapour at 1 atm	(R)	$\Delta S_{\text{system}} < 0$
(IV)	Adiabatic reversible compression of ideal gas	(S)	$\Delta S_{\text{surr.}} > 0$
		(T)	$W > 0$
		(U)	$\Delta H = 0$ or $\Delta H < 0$

Which of the following having incorrect combination considering List-I and List-II

 (A) I $\rightarrow$ P, Q, U

 (B) II $\rightarrow$ R,S,T

 (C) III $\rightarrow$ R,S,T

 (D) IV $\rightarrow$ Q,U

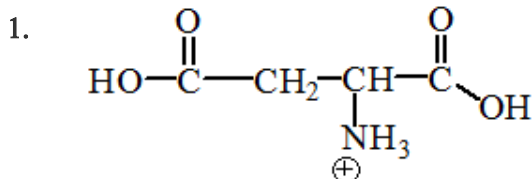
Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

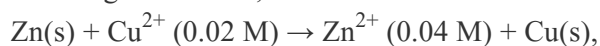
Zero Marks : 0 In all other cases.



$$pK_{a1} = 2 ; pK_{a2} = 4 ; pK_{a3} = 10$$

Find isoelectric point of the above amino acid.

2. For the galvanic cell,



$$E_{cell} = \text{_____} \times 10^{-2} \text{ V. (Nearest integer)}$$

$$[\text{Use : } E^0_{Cu/Cu^{2+}} = -0.34 \text{ V, } E^0_{Zn/Zn^{2+}} = +0.76 \text{ V, } \frac{2.303 RT}{F} = 0.059 \text{ V}]$$

3. A solution of 12.4 gm ethylene glycol in 100g H₂O is cooled to -5.58°C. What is the weight (in grams) of the ice separated from solution ($K_f(\text{H}_2\text{O}) = 1.86 \text{ K. kg mole}^{-1}$) ?

Space for Rough Work

4. A compound (X) on heating with an excess of NaOH solution gives gas (Y) which gives white fumes on exposure of HCl. Heating is continued to expel the gas completely. The resultant alkaline solution again liberates the same gas (Y) when heated with Zn powder. However, the compound (X) when heated alone does not give diatomic gas. If the number of atoms in one molecule of compound (X) is 'a' and the number of atoms in one molecule of gas (Y) is 'b' then find the value of a/b ?
5. Find the number of process in which hybridisation of underlined atom does not changed ?
- (i) $\underline{\text{B}}_2\text{H}_6 + \text{THF} \rightarrow$
- (ii) $\underline{\text{Cr}}\text{O}_4^{2-} + \text{H}^+ \rightarrow$
- (iii) $\text{K}_2[\underline{\text{Ni}}(\text{CN})_4] + \text{K} \xrightarrow{\text{NH}_3}$
- (iv) $\underline{\text{Si}}\text{O}_2 + \text{HF} \rightarrow$
- (v) $\text{PCl}_5 + \text{CH}_3\underline{\text{C}}\text{OOH} \rightarrow$
- (vi) $\text{KMnO}_4 + \text{Na}\underline{\text{N}}\text{O}_2 + \text{H}^+$
- (vii) $\text{H}_3\underline{\text{B}}\text{O}_3 + \text{NaOH} \rightarrow$
6. Consider following complexes for stereo isomerism only :
 $[\text{Co}(\text{NO}_2)_3(\text{NH}_3)_3]$, $[\text{Pt}(\text{en})\text{Cl}_2]$, $[\text{Co}(\text{en})_2(\text{NO}_2)\text{Cl}]$,
 $\text{Na}[\text{PtBrCl}(\text{NO}_2)(\text{NH}_3)]$, Sodium nitro prusside,
 $[\text{Ir}(\text{SCN})(\text{SO}_4)(\text{NH}_3)_4]$, $\text{Na}_3[\text{Cr}(\text{ox})_3]$
 Let number of complexes showing
 (i) Only geometrical isomerism = a
 (ii) Only optical isomerism = b
 (iii) Neither geometrical nor optical isomerism = c
 (iv) Showing both geometrical and optical isomerism = d
 Then find the value of $(a + b + d)/c$

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. In a triangle ABC points D, E lie on sides AC and AB respectively. M and N are mid points of side BD and CE respectively. If Δ_1 is area of triangle AMN, Δ_2 be the area of triangle ADE, Δ be area of triangle ABC, Δ_3 be area of quadrilateral BCDE, Δ_4 be area of quadrilateral MNDE. Which of the following is/are always true ?
 (A) $4\Delta_1 = \Delta_3$ (B) $4\Delta_1 + \Delta_2 = \Delta$ (C) $\Delta_1 = \Delta_4$ (D) $4\Delta_4 = \Delta_2 + \Delta$
2. Tangents are drawn from $\left(-\frac{\pi}{2}, 0\right)$ to $f: [0, \infty) \rightarrow \mathbb{R}; y = \sin x$ and $(a_i, \sin a_i)$ are the points of contact of these tangents for $i = 1, 2, 3, \dots$ and $a_{i+1} > a_i$. Choose the **correct** options(s)
 (A) $a_n + \frac{\pi}{2} = \tan a_n$ (B) $a_{n+1} - a_n < \pi$
 (C) $a_{n+2} - a_n > 2\pi$ (D) $\sin a_{2n} - \sin a_{2n-1} < 0$

Space for Rough Work

3. P is a point on ellipse $\frac{x^2}{36} + \frac{y^2}{20} = 1$ in first quadrant. $F_1(\alpha, 0)$ and $F_2(\beta, 0)$ are two foci of the ellipse ($\alpha < \beta$). Line PF_1 intersect circle $x^2 + y^2 - 6y + 8 = 0$ in Q. Choose the **correct** option(s)
- (A) The maximum value of $PQ + PF_2$, as P moves on the ellipse in first quadrant, is 8.
- (B) The minimum value of $PQ + PF_2$, as P moves on the ellipse in first quadrant, is 6.
- (C) If the line PF_1 is at maximum distance from (0, 3) then the value of $PQ + PF_2$ is $12 - \sqrt{24}$
- (D) The maximum value of $PQ + PF_2$, as P moves on the ellipse in first quadrant, is 12.
4. Let $n \geq 0$ be an integer and $I_n = \int_0^\pi \frac{1 - \cos nx}{1 - \cos x} dx$, then
- (A) $I_0, I_1, I_2, \dots$, is an A.P. (B) $I_0, I_1, I_2, \dots$, is a G.P.
- (C) $I_n = n\pi$; $n \geq 1$ (D) absolute value of the integrand is bounded
5. ABCD is a tetrahedron with $\triangle BCD$ being equilateral, having each side equal to 12 units. H is foot of perpendicular from A to $\triangle BCD$ and it lies inside the triangle if area of $\triangle CDH$ is three times the area of $\triangle BCH$ and area of $\triangle DBH$ is two times the area of $\triangle BCH$, $AH = 3$ and M is midpoint of side BD. Choose the **correct** option(s)
- (A) Perpendicular distance of Point H from side BC is $\sqrt{3}$
- (B) Perpendicular distance of Point H from side BD is $2\sqrt{3}$
- (C) Perpendicular distance of Point A from line CM is $\sqrt{13}$
- (D) Perpendicular distance of Point A from line CM is $\sqrt{15}$

Space for Rough Work

6. Tangent are drawn from $(-2, 0)$ to $y^2 = 8x$, radius of circle(s) that would touch these tangents and the corresponding chord of contact, can be equal to
 (A) $4(\sqrt{2} + 1)$ (B) $4(\sqrt{2} - 1)$ (C) $8\sqrt{2}$ (D) None of these
7. The equation of the plane which contains line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{4}$ and its projection in the plane $x + 2y + z = 6$, is/are
 (A) $x + 2y + z - 6 = 0$ (B) $9x - 2y - 5z + 4 = 0$
 (C) $x - 2y - z + 6 = 0$ (D) $9x + 2y - 5z + 4 = 0$
8. If A and B are any two different square matrices of order n with A - B is non-singular $A^3 = B^3$ and $A(A(B)) = B(B(A))$, then
 (A) $A^2 + B^2 = 0$ (B) $A^2 + B^2 = I$ (C) $A^2 + B^3 = I$ (D) $A^3 + B^3 = 0$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.

Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)

Negative Marks : -1 In all other cases

9. Match the List-I with List-II.

List-I		List-II	
(I)	$f: D_f \rightarrow \mathbb{R};$ $f(x) = (\sqrt{\tan x} + \sqrt{\sin x})(\sqrt{\tan x} - \sqrt{\sin x})$ Where D_f is domain of $f(x)$, then $f(x)$ is	(P)	An even function
(II)	$f: \mathbb{R} \rightarrow (1, 3]; f(x) = [x + 1] + [x - 1] - 2x + 3$, where $[.]$ denotes greatest integer function, then $f(x)$ is	(Q)	An odd function
(III)	$f: \mathbb{R} \rightarrow (-1, 1); f(x) = \frac{x}{1 + x }$; then $f(x)$ is	(R)	Injective function
(IV)	$f: \mathbb{R} \rightarrow [0, \infty); f(x) = \left(\ln \left(x + \sqrt{1 + x^2} \right) \right)^2$; then $f(x)$ is	(S)	Surjective function
		(T)	Periodic function
		(U)	Bijective function

Which of the following is correct combination ?

- (A) $I \rightarrow P, T$ (B) $II \rightarrow Q, T$ (C) $III \rightarrow Q, R, S, U$ (D) $III \rightarrow Q, R$

Space for Rough Work

10. Match the List-I with List-II.

List-I		List-II	
(I)	$f: D_f \rightarrow \mathbb{R};$ $f(x) = (\sqrt{\tan x} + \sqrt{\sin x})(\sqrt{\tan x} - \sqrt{\sin x})$ Where D_f is domain of $f(x)$, then $f(x)$ is	(P)	An even function
(II)	$f: \mathbb{R} \rightarrow (1, 3]; f(x) = [x + 1] + [x - 1] - 2x + 3$, where $[.]$ denotes greatest integer function, then $f(x)$ is	(Q)	An odd function
(III)	$f: \mathbb{R} \rightarrow (-1, 1); f(x) = \frac{x}{1 + x }$; then $f(x)$ is	(R)	Injective function
(IV)	$f: \mathbb{R} \rightarrow [0, \infty); f(x) = \left(\ln \left(x + \sqrt{1 + x^2} \right) \right)^2$; then $f(x)$ is	(S)	Surjective function
		(T)	Periodic function
		(U)	Bijective function

Which of the following is correct combination ?

(A) $IV \rightarrow P, S$

(B) $IV \rightarrow P, R, S$

(C) $II \rightarrow R, S, T, U$

(D) $II \rightarrow Q, R, S, U$

Space for Rough Work

11. Consider the system of equations.

$$a^2x + (a^2 - (b - c)^2)y + bcz = 0$$

$$b^2x + (b^2 - (c - a)^2)y + caz = 0$$

$$c^2x + (c^2 - (a - b)^2)y + abz = 0$$

List-I		List-II	
(I)	If $a + b + c = 0$ and no two of a, b, c are equal then	(P)	System of equation have unique solution
(II)	If $a = b = c = 0$ then	(Q)	System of equation have infinite number of solutions
(III)	If $a = b = c, a + b + c \neq 0$ then	(R)	$\forall x, y, z \in \mathbb{R}$, the system of equations will be true
(IV)	If $a + b + c \neq 0$ and no two out of a, b, c are equal, then	(S)	Solution of system of equations are in the form $(\lambda_1, \lambda_2, -\lambda_1 - \lambda_2)$, where $\lambda_1, \lambda_2 \in \mathbb{R}$
		(T)	System of equation have atleast two solutions
		(U)	System of equation have no solution

Which of the following is correct combination ?

(A) $I \rightarrow Q, R$

(B) $I \rightarrow Q, R, T$

(C) $II \rightarrow Q, R$

(D) $II \rightarrow Q, R, T$

Space for Rough Work

12. Consider the system of equations.

$$a^2x + (a^2 - (b - c)^2)y + bcz = 0$$

$$b^2x + (b^2 - (c - a)^2)y + caz = 0$$

$$c^2x + (c^2 - (a - b)^2)y + abz = 0$$

List-I		List-II	
(I)	If $a + b + c = 0$ and no two of a, b, c are equal then	(P)	System of equation have unique solution
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(III)	If $a = b = c, a + b + c \neq 0$ then	(R)	$\forall x, y, z \in \mathbb{R}$, the system of equations will be true
(IV)	If $a + b + c \neq 0$ and no two out of a, b, c are equal, then	(S)	Solution of system of equations are in the form $(\lambda_1, \lambda_2, -\lambda_1 - \lambda_2)$, where $\lambda_1, \lambda_2 \in \mathbb{R}$
		(T)	System of equation have atleast two solutions
		(U)	System of equation have no solution

Which of the following is correct combination ?

(A) III $\rightarrow$ Q

(B) III $\rightarrow$ Q,S,T

(C) IV $\rightarrow$ Q,T

(D) IV $\rightarrow$ U

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. Consider $f : (0, \infty) \rightarrow \mathbb{R}$, $f(x) = x^x$ be a differentiable function. Let $f(x) = k$ has two distinct real roots α and β such that $\max|\alpha - \beta| < \lambda$ then the least integral value of λ is
 2. If $\int_1^e \frac{1 - \ln x}{x\sqrt{x^2 + (\ln x)^2}} dx = \ln(k_1 + \sqrt{1 + e^2}) + k_2$, where k_1 and k_2 are integers; then $(k_1 - k_2)$ equals
 3. A question paper consists of 5 multiple choice questions with 4 options each. Out of these 3 questions have only one option correct and 1 mark will be awarded for correct answer while remaining 2 questions have one or more than one option(s) correct and two marks will be awarded for correct answer. The probability that student scored 4 marks in the paper; if he guesses all the answers (no negative marks will awarded for wrong answers), is $\frac{k}{100}$, then find k
-

Space for Rough Work

4. If m is the minimum value of $f(x)$, where $f(x) = x^{3/2} + \frac{1}{8x^{3/2}} - \frac{17}{4} \left(x + \frac{1}{4x} \right)$, $x > 0$, then the value of $|m|$ is
5. Let $p = \sum_{r=1}^{2021} (-1)^{r-1} \frac{{}^{2021}C_r}{2^{r+1} - 1}$ and $q = \sum_{r=1}^{\infty} \frac{(2^r - 1)^{2021}}{2^{2022r}}$, then the value of $p + q$ is
6. The quadratic equation $x^2 - (a + 5)x + 2a = 0$, $a \in I$, has rational roots α and β . Then the maximum value of $\left(\frac{\alpha^2 + \beta^2}{100} \right)$
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work